MEYER TURKU

SHIPYARD 1737

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SINCE AI CHALLENGE '26

Automated Class & Regulatory Compliance Checking

MEYER TURKU AT A GLANCE

We design and build cruise ships of the future.

1737
Building ships since

1970
World's first purpose-built cruise ship

17%
Global cruise-ship market

€2.14bn
2025 Turnover

~250 kGT/year
Production Capacity

2036
Extended Orderbook



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How can ship designs be checked against rules and regulations automatically?

CHALLENGE

Why Regulation and Compliance Checking?

1. Complex Regulatory Landscape

→ Many overlapping class rules, conventions and regulations

2. Manual & Time Consuming

→ Compliance checking requires engineering expertise and substantial effort

3. Earlier Detection & Less Downstream Impact

→ Identify compliance issues before they propagate through the design



Source: Viking Line

TASK DESCRIPTION

Develop and demonstrate an AI solution that checks ship design against rules and regulations

These rules and regulations can be for example:

SOLAS	International Convention for the Safety of Life at Sea
MARPOL	International Convention for the Prevention of Pollution from Ships
COLREG	International Regulations for Preventing Collisions at Sea
Classification Society Rules	e.g. DNV or Lloyds
PVAG	Passenger Vessel Accessibility Guidelines
ADA	Americans with Disabilities Act

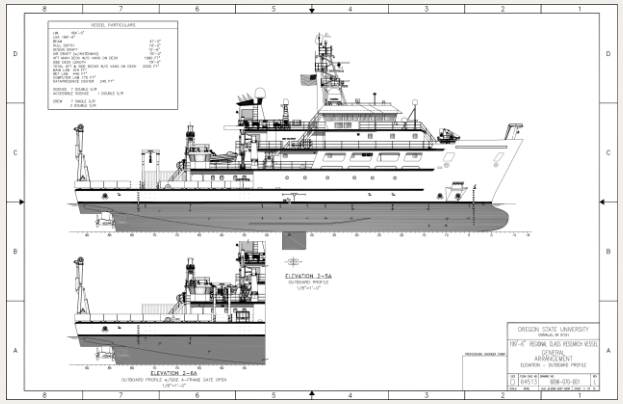
We are looking for a solution that assists our engineers in their day-to-day work and informs them of any issues found in the design



SHIP DESIGN

Oregon State University
Regional Class Research Vessel

→ Public Design of a current vessel under construction
→ Detail-Design level 3D model
→ General Arrangement and technical documentation
→ Links in the Since AI App.





Autodesk Navisworks is required to open the 3D model, student licenses from Autodesk are available via their website

You are not required to use the material provided by us. You can use one or several input sources.



Sources: Oregon State University, UNOLST, Glosten, Bollinger Shipyards

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What your demo should include.

Accuracy

→ Correctly identifies compliant vs. non-compliant design

→ Minimizes incorrect or misleading compliance findings

Can we trust the assessment?

Design Interpretation

→ Demonstrate that it can understand documents, drawings and/or 3D models

→ Clearly highlights why something is not compliant

Can it understand the design and show us why there is a problem?

Quality & Usability

→ Solid technical foundation

→ Usability in an engineering context

→ Clear demonstration of how the solution works

Is this a credible foundation for a useful engineering tool?

A robust foundation matters more than a polished UI

Point of Contact during the Hackathon



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Reach us on Discord & in Joki on Saturday

- Friday until **19:45**
- Saturday **08:00** – **19:00**
- Sunday from **08:00**

Rule / regulation	Availability	What does it cover?
SOLAS	Partially	International convention covering the safety of ships, including requirements intended to protect life at sea.
MARPOL	Partially	International convention focused on preventing pollution from ships.
COLREG	Yes	International rules governing ship navigation to prevent collisions at sea.
Load Line Convention	Partially	International convention establishing load-line requirements for ships.
Ballast Water Management Convention	Partially	Establishes requirements for the management of ships' ballast water and sediments.
Classification Society Rules, e.g. DNV	Yes	Technical classification rules and standards against which ship designs can be assessed.
PVAG (Passenger Vessel Accessibility Guidelines)	Yes	U.S. guidelines defining accessibility requirements for passenger vessels.
ADA (Americans with Disabilities Act)	Yes	U.S. accessibility requirements intended to ensure access for people with disabilities.